

Separation of Variables and Legendre Polynomial

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The Differential Operator under Different Coordinates

Formula 1. For Spherical: $d\ell = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin \theta d\phi \hat{\boldsymbol{\phi}}$ $d^3x = r^2 \sin \theta dr d\theta d\phi$

$$\text{Gradient : } \nabla V = \partial_r V \hat{\mathbf{r}} + \frac{1}{r} \partial_\theta V \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \partial_\phi V \hat{\boldsymbol{\phi}}$$

$$\text{Divergence : } \nabla \cdot \mathbf{E} = \frac{1}{r^2} \partial_r (r^2 E_r) + \frac{1}{r \sin \theta} \partial_\theta (\sin \theta E_\theta) + \frac{1}{r \sin \theta} \partial_\phi E_\phi$$

$$\begin{aligned} \text{Curl : } \nabla \times \mathbf{E} = & \frac{1}{r \sin \theta} [\partial_\theta (\sin \theta E_\phi) - \partial_\phi E_\theta] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \partial_\phi E_r - \partial_r (r E_\phi) \right] \hat{\boldsymbol{\theta}} \\ & + \frac{1}{r} [\partial_r (r E_\theta) - \partial_\theta E_r] \hat{\boldsymbol{\phi}} \end{aligned}$$

$$\text{Laplacian : } \nabla^2 V = \frac{1}{r^2} \partial_r (r^2 \partial_r V) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta V) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 V$$

Formula 2. For Cylindrical: $d\ell = dr \hat{\mathbf{r}} + r d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}$ $d^3x = r dr d\phi dz$

$$\text{Gradient : } \nabla V = \partial_r V \hat{\mathbf{r}} + \frac{1}{r} \partial_\phi V \hat{\boldsymbol{\phi}} + \partial_z V \hat{\mathbf{z}}$$

$$\text{Divergence : } \nabla \cdot \mathbf{E} = \frac{1}{r} \partial_r (r E_r) + \frac{1}{r} \partial_\phi E_\phi + \partial_z E_z$$

$$\begin{aligned} \text{Curl : } \nabla \times \mathbf{E} = & \left(\frac{1}{r} \partial_\phi E_z - \partial_z E_\phi \right) \hat{\mathbf{r}} + (\partial_z E_r - \partial_r E_z) \hat{\boldsymbol{\phi}} \\ & + \frac{1}{r} [\partial_r (r E_\phi) - \partial_\phi E_r] \hat{\mathbf{z}} \end{aligned}$$

$$\text{Laplacian : } \nabla^2 V = \frac{1}{r} \partial_r (r \partial_r V) + \frac{1}{r^2} \partial_\phi^2 V + \partial_z^2 V$$

1 Separation of Variables in Cartesian Coordinates

1.1 Separation of Variables as a Consequence of the Spectral Theorem

The Laplace equation gives

$$\nabla^2 V = 0 \quad (1)$$

within a region where the charge density is zero due to the Poisson equation. In the orthogonal coordinates (such as Cartesian coordinates and spherical coordinates), we can verify the property of self-adjointness in the Hilbert space. That is,

$$\langle f, Lg \rangle = \langle L^\dagger f, g \rangle \quad L^2(\Omega) = \left\{ f \mid \int_{\Omega} |f|^2 d^D x < \infty \right\}$$

with linear operator L and $f, g \in L^2(\Omega)$. In the well-defined real function, we define the inner product between f, g , that is

$$\langle f, g \rangle := \int_{\Omega} f^*(x)g(x) d^D x = \int_{\Omega} f(x)g(x) d^D x \quad (2)$$

Here, we focus on the one-dimensional $\Omega = [a, b]$, so $D = 1$. For example, if the linear operator $L = d/dx$, then

$$\langle f, Lg \rangle = \int_a^b f(x) \frac{dg}{dx} dx = f(x)g(x) \Big|_a^b - \int_a^b \frac{df}{dx} g dx \quad (3)$$

With the appropriate boundary condition(B.C.) $f(b)g(b) = f(a)g(a)$, we can obtain

$$\langle f, Lg \rangle = - \int_a^b \frac{df}{dx} g dx = \langle -Lf, g \rangle \equiv \langle L^\dagger f, g \rangle$$

In this case, $L^\dagger = -L$. **The term $L^\dagger$ depends on the different forms of L** ; in addition, whether it is self-adjoint or not depends on the different B.C. and domain. Furthermore, if two operators L_1, L_2 commute(if and only if), that is,

$$[L_1, L_2] = L_1 L_2 - L_2 L_1 = 0 \quad (4)$$

The spectral theorem states that

Theorem 1 (Spectral Theorem for Self-Adjoint Operators). *Let $\mathcal{H}$ be a Hilbert space and let $L : D(L) \subset \mathcal{H} \rightarrow \mathcal{H}$ be a densely defined self-adjoint linear operator. Then there exists a projection-valued measure $E(\lambda)$ on the real line such that*

$$L = \int_{\mathbb{R}} \lambda dE(\lambda).$$

Moreover, the following properties hold:

1. The spectrum of L is real. That is,

$$Lf = \lambda f \quad \lambda \in \mathbb{R}$$

2. Eigenfunctions corresponding to distinct eigenvalues are orthogonal. It can be denoted by

$$\lambda \neq \mu \Leftrightarrow \langle f_\lambda, f_\mu \rangle = 0$$

3. The Hilbert space $\mathcal{H}$ admits an orthogonal decomposition into spectral subspaces of L .

4. Any $f \in \mathcal{H}$ can be expanded in terms of the eigenfunctions of L (discrete spectrum) or via a spectral integral (continuous spectrum).

In Cartesian coordinates, the operator ∇^2 is

$$\nabla^2 = (\partial_x^2 + \partial_y^2 + \partial_z^2)$$

First, we can verify that the components of the operator are self-adjoint. To do so, we can write down

$$\begin{aligned} \langle f, \partial_x^2 g \rangle &= \int_a^b f^*(x) g''(x) dx \\ &= \left[f^*(x) g'(x) - f^{*'}(x) g(x) \right]_a^b + \int_a^b f^{*''}(x) g(x) dx \end{aligned}$$

With the following boundary conditions imposed on both f and g :

$$\begin{cases} \text{Dirichlet : } g(a) = g(b) = 0 \\ \text{Neumann : } g'(a) = g'(b) = 0 \end{cases} \quad (5)$$

The boundary term $\left[f^*(x) g'(x) - f^{*'}(x) g(x) \right]_a^b$ vanishes, so

$$\langle f, \partial_x^2 g \rangle = \langle \partial_x^2 f, g \rangle \quad (6)$$

So, the operators $\partial_x^2, \partial_y^2, \partial_z^2$ are self-adjoint.

Since the operators $\partial_x^2, \partial_y^2$, and ∂_z^2 are self-adjoint and mutually commuting, the spectral theorem guarantees the existence of a complete set of common eigenfunctions. Let $V(x, y, z)$ be such a common eigenfunction, satisfying

$$\partial_x^2 V = \lambda V, \quad \partial_y^2 V = \mu V, \quad \partial_z^2 V = \nu V \quad (7)$$

Consider the first equation. For each fixed y and z , the function $V(\cdot, y, z)$ on the Hilbert space $L^2(\Omega_x)$ with the same B.C. satisfies a one-dimensional eigenvalue problem for ∂_x^2 . Hence, it can be written as

$$V(x, y, z) = X(x) F(y, z). \quad (8)$$

Substituting this form into the second equation yields

$$\partial_y^2 [X(x) F(y, z)] = X(x) \partial_y^2 F(y, z) = \mu X(x) F(y, z) \quad (9)$$

which implies

$$F(y, z) = Y(y) Z(z) \quad (10)$$

Therefore, the common eigenfunctions necessarily factorize as

$$V(x, y, z) = X(x) Y(y) Z(z) \quad (11)$$

In the product Hilbert space

$$L^2(\Omega_x \times \Omega_y \times \Omega_z) \simeq \bigotimes_{i=x,y,z} L^2(\Omega_i) \quad (12)$$

These product functions form a complete orthogonal basis. Consequently, general solutions can be expressed as linear combinations of such separable eigenfunctions. So, we can rewrite the original problem as

$$(\partial_x^2 + \partial_y^2 + \partial_z^2)V = (\lambda + \mu + \nu)V = 0 \quad (13)$$

with

$$\frac{X''(x)}{X(x)} = \lambda, \quad \frac{Y''(y)}{Y(y)} = \mu, \quad \frac{Z''(z)}{Z(z)} = \nu \quad (14)$$

Hence, the separation constants arise as eigenvalues of the commuting self-adjoint operators. In the Laplace equation, it satisfies

$$\lambda + \mu + \nu = 0 \quad (15)$$

1.2 2-D Problem

In Cartesian coordinates, 2-D problems mean that the potential V is independent of z (or x, y). The B.C. satisfies

1. $V = 0$ When $y = 0$,
2. $V = 0$ When $y = a$,
3. $V = V_0(y)$ When $x = 0$,
4. $V \rightarrow 0$ as $x \rightarrow \infty$

We assume that the potential is $V(x, y) = X(x)Y(y)$, the Laplacian yields

$$\nabla^2 V(x, y) = \frac{\partial^2 V}{\partial^2 x} + \frac{\partial^2 V}{\partial^2 y} = 0$$

By (14) and (15), this two dimensional PDE can be written as

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = \lambda + \mu = 0 \Rightarrow \begin{cases} \frac{X''(x)}{X(x)} = \lambda \\ \frac{Y''(y)}{Y(y)} = \mu \end{cases}$$

so $\mu = -\lambda$. Then the solutions to the functions are

$$X(x) = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}, \quad Y(y) = C \sin(\sqrt{\lambda}y) + D \cos(\sqrt{\lambda}y) \quad (16)$$

By condition 1, $V(x, 0) = X(x)Y(0) = 0$, then $D = 0$; by condition 2.

$$V(x, a) = X(x)Y(a) = X(x)[C \sin(\sqrt{\lambda}a)] = 0$$

We've known that $\sin(n\pi) = 0$, $n \in \mathbb{N}$, so $\sqrt{\lambda} = n\pi/a$, $n \in \mathbb{N}$.

By condition 4, $V(\infty, y) \rightarrow 0$, that is

$$\lim_{x \rightarrow \infty} V(x, y) = \left(Ae^{n\pi x/a} \Big|_{x \rightarrow \infty} + Be^{-n\pi x/a} \Big|_{x \rightarrow \infty} \right) C \sin\left(\frac{n\pi y}{a}\right) = 0$$

We obtain $A = 0$. Let $BC = k$, and note that not only one solution V satisfies this PDE. If V_i all satisfy the B.C., then

$$\nabla^2 V = \sum \alpha_i \nabla^2 V_i = 0 \quad (17)$$

So the solution $V(x, y)$ can be written as

$$V(x, y) = \sum_{n=1}^{\infty} k_n e^{-n\pi x/a} \sin\left(\frac{n\pi y}{a}\right) \quad (18)$$

Finally, condition 4 satisfies

$$V_0(y) = V(0, y) = \sum_{n=1}^{\infty} k_n \sin\left(\frac{n\pi y}{a}\right)$$

We multiply both sides by $\sin(m\pi y/a)$ and integrate them in the range of y ; this is called **Fourier's Trick**, which is a fucking good technique. We obtain

$$\int_0^a V_0(y) \sin\left(\frac{m\pi y}{a}\right) dy = \sum_{n=1}^{\infty} k_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy \quad (19)$$

By the spectral theorem, the integral

$$\int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy = \frac{a}{2} \delta_{nm} \quad (20)$$

$a/2$ is independent of the number n , so the constant k_n is

$$k_n = \frac{2}{a} \int_0^a V_0(y) \sin\left(\frac{n\pi y}{a}\right) dy \quad (21)$$

Suppose that $V_0(y) = V_0$ as it's a metal plate with a constant potential. Then the integral gives

$$k_n = \frac{2V_0}{a} \int_0^a \sin\left(\frac{n\pi y}{a}\right) dy = \frac{2V_0}{n\pi} (1 - \cos n\pi) = \begin{cases} 0, & \text{even } n \\ \frac{4V_0}{n\pi}, & \text{odd } n \end{cases} \quad (22)$$

Substituting (22) into (18), we can obtain the final solution to the PDE:

$$V(x, y) = \frac{4V_0}{\pi} \sum_{\text{odd } n} \frac{1}{n} e^{-n\pi x/a} \sin\left(\frac{n\pi y}{a}\right) \quad (23)$$

1.3 3-D Problem

When we consider the 3-D problem, we need more boundary conditions to determine the solution $V(x, y, z)$ to the PDE. We have six B.C.:

1. $V = 0$ when $y = 0$,
2. $V = 0$ when $y = a$,
3. $V = 0$ when $z = 0$,
4. $V = V_0$ when $z = b$,
5. $V = 0$ when $x = 0$,
6. $V = 0$ when $x = c$.

In this case, we express $V(x, y, z)$ to $V = X(x)Y(y)Z(z)$, the Laplacian yields

$$\nabla^2 V(x, y, z) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0$$

By (14) and (15), this three dimensional PDE can be written as

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} + \frac{Z''(z)}{Z(z)} = -\lambda - \mu + \nu = 0 \Rightarrow \begin{cases} \frac{X''(x)}{X(x)} = -\lambda \\ \frac{Y''(y)}{Y(y)} = -\mu \\ \frac{Z''(z)}{Z(z)} = \nu \end{cases}$$

so $\lambda + \mu = \nu$.¹ Then the solutions to the functions are

$$\begin{cases} X(x) = A \sin(\sqrt{\lambda}x) + B \cos(\sqrt{\lambda}x) \\ Y(y) = C \sin(\sqrt{\mu}y) + D \cos(\sqrt{\mu}y) \\ Z(z) = E e^{\sqrt{\lambda+\mu}z} + F e^{-\sqrt{\lambda+\mu}z} \end{cases} \quad (24)$$

By the boundary conditions, we obtain

$$\begin{cases} B = 0 \\ D = 0 \\ E + F = 0 \end{cases} \quad \begin{cases} \sqrt{\lambda} = \frac{n\pi}{c}, n \in \mathbb{N} \\ \sqrt{\mu} = \frac{m\pi}{a}, m \in \mathbb{N} \end{cases}$$

Substituting these into (24), the potential can be written as

$$\begin{aligned} V(x, y, z) &= \sum_{m,n=1}^{\infty} k_{nm} \left(e^{\sqrt{(n\pi/c)^2 + (m\pi/a)^2}z} - e^{-\sqrt{(n\pi/c)^2 + (m\pi/a)^2}z} \right) \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) \\ &= \sum_{m,n=1}^{\infty} 2k_{nm} \left[\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}z\right) \right] \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) \end{aligned} \quad (25)$$

When $z = b$, $V(x, y, b) = V_0$, and we have

$$V_0 = \sum_{m,n=1}^{\infty} 2k_{nm} \left[\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right) \right] \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) \quad (26)$$

¹Since λ & μ are constant in $\mathbb{R}$, whether they are positive or not are all equivalent.

Use Fourier's trick, multiply both sides by $\sin(n'\pi x/c) \sin(m'\pi y/a)$, then

$$\begin{aligned}
& V_0 \int_0^a \int_0^c \sin\left(\frac{n'\pi x}{c}\right) \sin\left(\frac{m'\pi y}{a}\right) dx dy \\
&= \sum_{m,n=1}^{\infty} 2k_{nm} \left[\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right) \right] \\
& \quad \int_0^c \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{n'\pi x}{c}\right) dx \int_0^a \sin\left(\frac{m\pi y}{a}\right) \sin\left(\frac{m'\pi y}{a}\right) dy \\
&= \sum_{m,n=1}^{\infty} 2k_{nm} \left[\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right) \right] \frac{ac}{4} \delta_{nn'} \delta_{mm'}
\end{aligned}$$

Then the constant k_{nm} is:

$$\begin{aligned}
k_{nm} &= \frac{2V_0}{ac \sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right)} \int_0^a \int_0^c \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right) dx dy \\
&= \frac{2V_0}{nm\pi^2} \frac{(1 - \cos n\pi)(1 - \cos m\pi)}{\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right)} = \begin{cases} 0, & \text{even } n, m \\ \frac{8V_0}{nm\pi^2 \sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right)} & \text{odd } n, m \end{cases} \quad (27)
\end{aligned}$$

Substituting this into (25), we can therefore obtain

$$\boxed{V(x, y, z) = \frac{8V_0}{\pi^2} \sum_{\text{odd } n, m} \frac{1}{nm} \frac{\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}z\right)}{\sinh\left(\pi\sqrt{(n/c)^2 + (m/a)^2}b\right)} \sin\left(\frac{n\pi x}{c}\right) \sin\left(\frac{m\pi y}{a}\right)} \quad (28)$$

Spectral origin of admissible separated solutions

Consider the Laplace equation on a rectangular domain

$$\Omega = (0, c) \times (0, a) \times (0, b), \quad \nabla^2 V = 0 \text{ in } \Omega,$$

subject to Dirichlet boundary conditions

$$V = 0 \quad \text{on } x = 0, c; \ y = 0, a; \ z = 0, \quad V = V_0 \quad \text{on } z = b$$

The purpose of the separation of variables is not to construct a single solution, but to expand the solution on a basis compatible with the boundary conditions, so that arbitrary boundary data can be represented. Let us assume a separated form

$$V(x, y, z) = X(x)Y(y)Z(z)$$

Substitution into the Laplace equation yields

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0.$$

This equality implies that each term must be equal to a separation constant, leading to ordinary differential equations for each coordinate. However, the admissible forms of X and Y are not arbitrary. Due to the homogeneous Dirichlet boundary conditions

$$X(0) = X(c) = 0, \quad Y(0) = Y(a) = 0,$$

the functions X and Y must belong to the domain of the self-adjoint operators

$$L_x = -\frac{d^2}{dx^2}, \quad L_y = -\frac{d^2}{dy^2}$$

defined on $L^2(0, c)$ and $L^2(0, a)$ respectively, with Dirichlet boundary conditions. By Sturm–Liouville theory, these operators admit discrete spectra

$$\lambda_n = \left(\frac{n\pi}{c}\right)^2, \quad \mu_m = \left(\frac{m\pi}{a}\right)^2, \quad m, n \in \mathbb{N}$$

with corresponding eigenfunctions

$$X_n(x) = \sin\left(\frac{n\pi x}{c}\right), \quad Y_m(y) = \sin\left(\frac{m\pi y}{a}\right)$$

These eigenfunctions form complete orthogonal bases of their respective Hilbert spaces. Consequently, any admissible solution must be expandable as

$$V(x, y, z) = \sum_{m,n} A_{mn} X_n(x) Y_m(y) Z_{mn}(z),$$

where each Z_{mn} satisfies

$$Z_{mn}'' - \nu_{mn}^2 Z_{mn} = 0, \quad \nu_{mn} = \sqrt{\lambda_m + \mu_n}.$$

The choice of hyperbolic functions in the z -direction arises naturally, since the boundary conditions

$$Z_{mn}(0) = 0, \quad Z_{mn}(b) \neq 0$$

do not define an eigenvalue problem, but instead determine the amplitude of each mode required to match the prescribed boundary data.

In contrast, assigning hyperbolic functions to X or Y would be incompatible with the Dirichlet boundary conditions on finite intervals. Such functions do not yield nontrivial discrete spectra and therefore cannot form a complete orthogonal basis. Hence, they are excluded by the spectral structure of the underlying operators.

In summary, the selection of separated solutions is dictated not by the Laplace equation itself, but by the spectral properties of the self-adjoint operators associated with each coordinate and their boundary conditions.

2 Separation of Variables in Spherical Coordinates

With a similar concept to the Laplace equation in orthogonal coordinates, the Laplacian in spherical coordinates can be rewritten as

$$\nabla^2 = \frac{1}{r^2} \left[\underbrace{\partial_r (r^2 \partial_r)}_{L_r} + \underbrace{\Delta_{\theta\phi}}_{\tilde{L}_{\theta\phi}} \right] \quad (29)$$

where

$$\Delta_{\theta\phi} = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2$$

Since L_r and $\Delta_{\theta\phi}$ are self-adjoint with some boundary conditions, that is

$$\langle f, L_r g \rangle = \langle L_r^\dagger f, g \rangle, \quad \langle f, \Delta_{\theta\phi} g \rangle = \langle \Delta_{\theta\phi}^\dagger f, g \rangle$$

At the same time, they commute as linear operators, i.e.

$$[L_r, \Delta_{\theta\phi}] = 0$$

With the two properties, like Section 1, we can further separate the variable of the solution to

$$V(r, \theta, \phi) = f(r) A_{\theta\phi}(\theta, \phi) \quad (30)$$

Besides, the operator $\Delta_{\theta\phi}$ can also be written as

$$\Delta_{\theta\phi} = \frac{1}{\sin^2 \theta} \left[\underbrace{\sin \theta \partial_\theta (\sin \theta \partial_\theta)}_{L_\theta} + \underbrace{\partial_\phi^2}_{\tilde{L}_\phi} \right]$$

Similarly, we can also separate the function $A_{\theta\phi}(\theta, \phi)$ into

$$A_{\theta,\phi} = \Theta(\theta) \Phi(\phi) \quad (31)$$

Combining (30) and (31), we obtain

$$V(r, \theta, \phi) = f(r)\Theta(\theta)\Phi(\phi) \quad (32)$$

where we define $f(r) := R(r)/r$. The Laplace equation $\nabla^2 V = 0$ under the spherical coordinates gives

$$\nabla^2 V = \underbrace{\Theta\Phi \frac{d^2 R}{dr^2} + \frac{R\Phi}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{R\Theta}{r^2 \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2}}_{\times \frac{r^2 \sin^2 \theta}{R\Theta\Phi}} = 0 \quad (33)$$

$$\Rightarrow r^2 \sin^2 \theta \left[\frac{1}{R} \frac{d^2 R}{dr^2} + \frac{1}{\Theta r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) \right] + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = 0$$

Note that the last term Φ is independent of the equation, i.e., is isolated from the equation. So, it must be a const, which is

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -k^2 \quad \Phi = C_{\pm} e^{\pm ik\phi}$$

Substitute this into (33), we obtain

$$\frac{r^2}{R} \frac{d^2 R}{dr^2} + \left[\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{k^2}{\sin^2 \theta} \right] = 0$$

We assume that the first term on the left is λ , and the other must be $-\lambda$. This yields to

$$\begin{cases} \frac{r^2}{R} \frac{d^2 R}{dr^2} = \lambda \\ \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{k^2}{\sin^2 \theta} = -\lambda \end{cases} \quad (34)$$

The equation at the bottom of (34) can be written as

$$\frac{d}{\sin \theta d\theta} \left(\sin^2 \theta \frac{d\Theta}{\sin \theta d\theta} \right) + \left[\lambda - \frac{k^2}{\sin^2 \theta} \right] \Theta = 0 \quad (35)$$

2.1 The Convergence & Divergence of the Polynomial Solution

In such an analysis, we shall simplify it to be an azimuthally symmetric problem cause we can rewrite Eq.(35) as a Legendre equation², i.e., $k^2 = 0$.

By Eq.(35), let $x = \cos \theta$, $dx = -\sin \theta d\theta$. Thus Eq.(35) becomes

$$\frac{d}{dx} \left[(1-x^2) \frac{d\Theta}{dx} \right] + \lambda \Theta = 0 \quad (36)$$

²Legendre Equation: $(1-x^2)y'' - 2xy' + \lambda y = 0$

and WLOG, Θ can be considered a function with respect to x .³ We consider the whole range of the colatitudinal direction, i.e., $0 \leq \theta \leq \pi$, and $-1 \leq x \leq 1$. We consider the solution to be represented by a power series of the form

$$\Theta(x) = x^\alpha \sum_{j=0}^{\infty} a_j x^j \quad (37)$$

where α is a parameter to be determined. (It must have some conditions) Substitute (37) into (36), we obtain

$$\sum_{j=0}^{\infty} \left\{ (\alpha + j)(\alpha + j - 1) a_j x^{\alpha+j-2} - [(\alpha + j)(\alpha + j + 1) - \lambda] a_j x^{\alpha+j} \right\} = 0 \quad (38)$$

The lowest power of x when $j = 0$ is $x^{\alpha-2}$. The term of lowest order in x plays a special role. Since no other term in the series carries the same power, it cannot be cancelled by contributions from higher-order terms. Therefore, its coefficient must vanish independently, which yields the indicial equation.

$$\alpha(\alpha - 1) = 0 \Rightarrow \alpha = 1, 0 \quad (39)$$

By (38), we can find out the recurrence relation that

$$a_{j+2} = \left[\frac{(\alpha + j)(\alpha + 1 + j) - \lambda}{(\alpha + j + 1)(\alpha + j + 2)} \right] a_j \quad (40)$$

Note that from this recurrence relation, we can find that the series will diverge when the coefficients don't terminate, which means that when $j \rightarrow \infty$, it yields

$$\lim_{j \rightarrow \infty} \frac{a_{j+2}}{a_j} = \lim_{j \rightarrow \infty} \frac{(\alpha + j)(\alpha + 1 + j) - \lambda}{(\alpha + j + 1)(\alpha + j + 2)} = \frac{j^2}{j^2} = 1 \quad (41)$$

So the power series

$$\Theta(x) = x^\alpha \sum_j x^j = \frac{x^\alpha}{1 - x}, \quad -1^+ \leq x \leq 1^- \quad (42)$$

It is obvious that the function must "explode" on the boundaries $x = \pm 1$. That is to say, somehow, the series will definitely terminate at one of the terms. This indicates that

- $\alpha = 0$, $\exists j = l \in \mathbb{R}$ such that $j(j + 1) - \lambda = l(l + 1) - \lambda = 0$.
- $\alpha = 1$, $\exists j - 1 = l \in \mathbb{R}$ such that $[1 + (j - 1)][(j - 1) + 1] - \lambda = l(l + 1) - \lambda = 0$.

³Since the θ equation for $\Theta(\theta)$ is expressed in terms of x

So, the constant λ must satisfy the specific form, which is

$$\lambda = l(l + 1) \quad (43)$$

By convention, these polynomials are normalized to have the value unity at $x = 1$ and are called the *Legendre polynomials of order l* , i.e., $P_l(x)$, which is denoted by the *Rodrigue's formula*:

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l \quad (44)$$

2.2 The Form of General Solution to $R(r)$

We've known that when we want to terminate the series of the solution, the constant λ must have the specific form $l(l + 1)$. Substituting $\lambda = l(l + 1)$ into (34), we have

$$r^2 \frac{d^2 R}{dr^2} - l(l + 1)R = 0 \quad (45)$$

This is a Cauchy equation (We can let $r = e^z$ but I don't give a fuck). Let the operator $D = r \cdot d/dr$, then D^2 equals to

$$D^2 = r \frac{d}{dr} \left(r \frac{d}{dr} \right) = r \frac{d}{dr} + r^2 \frac{d}{dr^2} \quad (46)$$

So (45) becomes

$$[D^2 - D - l(l + 1)]R = (D + l)[D - (l + 1)]R = 0 \quad (47)$$

Let $u = [D - (l + 1)]R$, then

$$\begin{cases} ru' + lu = 0 \Rightarrow \frac{du}{dr} + \frac{l}{r}u = 0 \Rightarrow u = Cr^{-l} \\ rR' - (l + 1)R = Cr^{-l} \Rightarrow \frac{dR}{dr} - \frac{(l + 1)}{r}R = Cr^{-l-1} \end{cases}$$

Then the final solution to $R(r)$ is

$$R(r) = Ar^{l+1} + \frac{B}{r^l} \quad (48)$$

Note that the main function we separated in the beginning is $f(r) = R/r$, so the final solution is

$$f(r) = Ar^l + \frac{B}{r^{l+1}} \quad (49)$$

2.3 The Orthogonality of the Legendre Polynomial

Here, I want to use another method to prove the orthogonality of the Legendre Polynomial. Since we've known that the form of $\Theta(x)$ with polynomials is Legendre polynomial. Substituting (44) into the differential equation (36). Assume that there is another polynomials $P_n(x)$, and we multiply both sides by $P_n(x)$ and integrate both sides, which yields

$$\int_{-1}^1 P_n(x) \left[\frac{d}{dx}(1-x^2) \frac{dP_l(x)}{dx} + l(l+1)P_l(x) \right] dx = 0 \quad (50)$$

This can be simplified to

$$\begin{aligned} & \int_{-1}^1 P_n(x) \left[\frac{d}{dx}(1-x^2) \frac{dP_l(x)}{dx} + l(l+1)P_l(x) \right] dx \\ &= P_n(x)(1-x^2) \frac{dP_l(x)}{dx} \Big|_{-1}^1 - \int_{-1}^1 (1-x^2) \frac{dP_n(x)}{dx} \frac{dP_l(x)}{dx} dx + \int_{-1}^1 l(l+1)P_n(x)P_l(x) dx \\ &= \int_{-1}^1 l(l+1)P_n(x)P_l(x) dx - \int_{-1}^1 (1-x^2) \frac{dP_n(x)}{dx} \frac{dP_l(x)}{dx} dx \end{aligned} \quad (51)$$

Interchange P_l and P_n , we have

$$\int_{-1}^1 n(n+1)P_n(x)P_l(x) dx - \int_{-1}^1 (1-x^2) \frac{dP_n(x)}{dx} \frac{dP_l(x)}{dx} dx = 0 \quad (52)$$

Subtract the two equations (51) & (52) to obtain

$$[l(l+1) - n(n+1)] \int_{-1}^1 P_n(x)P_l(x) dx = 0$$

If $l \neq n$, then we can verify that

$$\int_{-1}^1 P_n(x)P_l(x) dx = 0 \quad (53)$$

If $l = n$, the integral becomes

$$\int_{-1}^1 [P_l(x)]^2 dx = \frac{1}{2^{2l}(l!)^2} \int_{-1}^1 \frac{d^l}{dx^l}(x^2-1)^l \frac{d^l}{dx^l}(x^2-1)^l dx = 0 \quad (54)$$

We can apply integration by parts l times to get

$$\begin{aligned}\int_{-1}^1 \frac{d^l}{dx^l}(x^2-1)^l \frac{d^l}{dx^l}(x^2-1)^l dx &= \frac{d^{l-1}}{dx^{l-1}}(x^2-1)^l \frac{d^l}{dx^l}(x^2-1)^l \Big|_{-1}^1 - \int_{-1}^1 (\dots) dx \\ &= \int_{-1}^1 \frac{d^{l-1}}{dx^{l-1}}(x^2-1)^l \frac{d^{l+1}}{dx^{l+1}}(x^2-1)^l dx = \dots \\ &= (-1)^l \int_{-1}^1 (x^2-1)^l \frac{d^{2l}}{dx^{2l}}(x^2-1)^l dx\end{aligned}$$

Differentiate $(x^2-1)^l$ for $2l$ times, we obtain

$$\frac{d^{2l}}{dx^{2l}}(x^2-1)^l = \frac{d^{2l}}{dx^{2l}} \sum_{n=0}^l \binom{l}{n} x^{2n} (-1)^{l-n} = (2l)!$$

Substituting the results above into (54), it becomes

$$\int_{-1}^1 [P_l(x)]^2 dx = \frac{(2l)!}{2^{2l}(l!)^2} \int_{-1}^1 (1-x^2)^l dx \quad (55)$$

Note that $(1-x^2)^l$ is an even function, and it is symmetric on the interval $[-1, 1]$. Let $t = x^2$, $dt = 2x dx$, then $dx = 1/2 t^{-1/2} dt$. With these conditions, we can rewrite (55) into

$$\int_{-1}^1 [P_l(x)]^2 dx = \frac{(2l)!}{2^{2l}(l!)^2} \cdot 2 \int_0^1 \frac{1}{2} t^{-1/2} (1-t)^l dt = \frac{(2l)!}{2^{2l}(l!)^2} B\left(\frac{1}{2}, l+1\right) \quad (56)$$

The beta function $B(1/2, l+1)$ is

$$B\left(\frac{1}{2}, l+1\right) = \frac{\Gamma\left(\frac{1}{2}\right) \Gamma(l+1)}{\Gamma\left(l+\frac{3}{2}\right)} = \frac{\sqrt{\pi} \cdot l!}{\frac{(2l+1)!}{2^{2l+1}(l!)}} = \frac{2^{2l+1}(l!)^2}{(2l+1)!}$$

This yields

$$\int_{-1}^1 [P_l(x)]^2 dx = \frac{(2l)!}{2^{2l}(l!)^2} \frac{2^{2l+1}(l!)^2}{(2l+1)!} = \frac{2}{2l+1} \quad (57)$$

The results of (53) and (57) truly show the orthogonality of the polynomials, that is

$$\boxed{\int_{-1}^1 P_n(x) P_l(x) dx = \frac{2}{2l+1} \delta_{nl}} \quad (58)$$

This property can help us to solve the PDE in the spherical coordinates when we solve the potential problem in Electrodynamics.

2.4 General Solution with Symmetric B.C.

By 2.1 and 2.2, the general solution to the potential $V(r, \theta, \phi)$, which is independent of azimuthal components, can be denoted by

$$V(r, \theta, \phi) = f(r)\Theta(\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta) \quad (59)$$

where P_l is the Legendre polynomial.

1. The potential $V_0(\theta)$ is specified on the surface of a hollow sphere, of radius R . Find the potential $V(r, \theta)$ **inside** the sphere.

Ans: The general solution is

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos \theta)$$

In this case, $B = 0$, or it will bangbang at the origin of the sphere. At $r = R$, we get

$$V(R, \theta) = \sum_{l=0}^{\infty} A_l R^l P_l(\cos \theta) = V_0(\theta) \quad (60)$$

Multiply both sides by $P_l(\cos \theta)$ and integrate them, we can obtain

$$\left\{ \begin{array}{l} \sum_{l=0}^{\infty} A_l R^l \int_0^{\pi} P_l(\cos \theta) P_l(\cos \theta) \sin \theta d\theta = \sum_{l=0}^{\infty} A_l \frac{2R^l}{2l+1} \\ \int_0^{\pi} V_0(\theta) P_l(\theta) \sin \theta d\theta \end{array} \right.$$

So the coefficient A_l is

$$A_l = \frac{2l+1}{2R^l} \int_0^{\pi} V_0(\theta) P_l(\theta) \sin \theta d\theta \quad (61)$$

Substituting (61) into $V(r, \theta)$, the final solution to the PDE is

$$V(r, \theta) = \sum_{l=0}^{\infty} \left[\frac{(2l+1)r^l}{2R^l} \int_0^{\pi} V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \right] P_l(\cos \theta) \quad (62)$$

Note that how many terms we need to take is due to the form of the potential on the surface. For example, when $V_0(\theta) = k[\alpha_0 P_0(\cos \theta) + \dots + \alpha_n P_n(\cos \theta)]$, then we only need to take n terms.

2. The potential $V_0(\theta)$ is specified on the surface of a hollow sphere, of radius R . Find the potential $V(r, \theta)$ **outside** the sphere.

Ans: In this case, $A = 0$, or it will bangbang at the infinity distant from the sphere. At $r = R$, we get

$$V(R, \theta) = \sum_{l=0}^{\infty} \frac{B_l}{R^{l+1}} P_l(\cos \theta) = V_0(\theta)$$

The coefficient B_l is

$$B_l = \frac{(2l+1)R^{l+1}}{2} \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta$$

Then the final solution is

$$V(r, \theta) = \sum_{l=0}^{\infty} \left[\frac{(2l+1)R^{l+1}}{2r^{l+1}} \int_0^\pi V_0(\theta) P_l(\cos \theta) \sin \theta d\theta \right] P_l(\cos \theta) \quad (63)$$